Reg. No. | | | | | | | | | |

Question Paper Code : 14649

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2018

Third Semester

Common to B.E. – Computer Science and Engineering & B.Tech – Information Technology

15CS302 – DIGITAL PRINCIPLES AND SYSTEM DESIGN

(Regulations: Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. The simplified SOP (Sum Of Product) form of the boolean expression (P + Q' + R') . (P + Q' + R) . (P + Q + R') is _____.

   A) (P'.Q + R') B) (P + Q'.R') C) (P'.Q + R) D) (P.Q + R)

2. _____ and _____ are the 9's and 10's complements of 24,681,234.

3. Parity Checkers are built by using

   A) AND gates B) OR gates

   C) NAND gates D) XOR gates

4. How can a decoder become a demultiplexer? Explain with example.

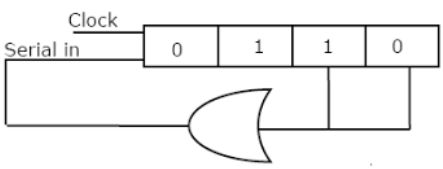
5. _____ number of address lines and ____ number of data lines are needed for a 32K x 8 RAM chip.

6. Is it advantageous to use coincident decoding for a 2K x 8 RAM? Justify.

7. The initial content of a 4 bit serial in serial out right shift register is shown in the figure below. What will be the contents of the shift register after 3 clock pulses?

   A) 0000 B) 1110

   C) 1010 D) 1111



8. How can a D flip-flop be converted into a T flip-flop?

9. Differentiate static-0 and static-1 hazards.

10. What are the different types of races?

PART B – (5 × 16 = 80 Marks)

11. a) i. Find the minimal sum of products for the Boolean expression, $F=\sum_m (1,2,3,5,7,8,9,10,11,12)+ \sum_d (14,15)$. Also draw the circuit using NAND logic. (12 Marks)

11. a) ii. Describe the canonical forms and standard forms of boolean functions with examples. (4 Marks)

OR

11. b) i. Find the minimal sum of products for the Boolean expression, $F=\sum_m (1,2,4,7,8,10,11,12,14,15)+ \sum_d(0,5)$, using the Quine-McCluskey method. (12 Marks)

11. b) ii. State and prove DeMorgan's law and Distributive law. (4 Marks)

12. a) i. Implement the following Boolean function using 8 × 1 multiplexer:
F(A,B,C,D)=A'BD'+ACD+B'CD+A'B'CD+A'BC'D+A'B'C'D (10 Marks)

12. a) ii. Construct a 5 to 32 line decoder with four 3 to 8 line decoders with enable and a 2 to 4 line decoder. (6 Marks)

OR

12. b) i. Design a code converter that converts a decimal digit from 2421 code to BCD. (10 Marks)

12. b) ii. What is a full adder? Design a full adder using multiplexers. (6 Marks)

13. a) i. Implement the functions: X = A'BC + ABC + A'B'C' and Y = ABC + AB'C using a PLA. (10 Marks)

13. a) ii. Explain the process of memory decoding. Also discuss how multiplexing makes the memory design efficient. (6 Marks)

OR

13. b) i. Implement the following 3 Boolean functions using PAL:

F1(A,B,C)=Σ(2,3,5,7)

F2(A,B,C)=Σ(0,1,5)

F3(A,B,C)=Σ(0,2,3,5) (10 Marks)

13. b) ii. Describe the different methods of ROM programming and the types of ROM. (6 Marks)

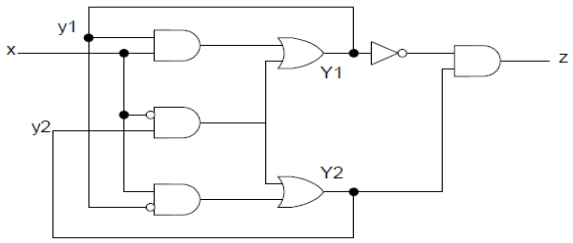
14. a) List out the steps for designing a synchronous sequential Circuit. Also design a synchronous sequential circuit with JK flip flops having one input and one output. The output of the circuit is 1 whenever a sequence 010 is received, otherwise the output is 0. (16 Marks)

OR

14. b) i. Design a synchronous BCD counter. Draw the necessary state diagram, state table and K maps. Design the circuit using JK flip flops. (12 Marks)

14. b) ii. Write a VHDL code for a modulo 10 counter. (4 Marks)

15. a) i. Analyze the following asynchronous sequential circuit by constructing transition table and flow table.



(8 Marks)

15. a) ii. Find a circuit that has no static hazards and implements the boolean function F(A,B,C,D)=∑(0, 2, 6, 7, 8, 10, 12). (8 Marks)

OR

15. b) A circuit has two inputs T and C and output Q. The output state is complemented if T = 1 and C goes from 1 to 0. Otherwise Q remains same. Design the asynchronous circuit. (16 Marks)